

IASI, Romania

WMO: 150900

Lat: 47.1636N

Lon: 27.6273E

Elev: 76

StdP: 100.42

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.6	-12.4	-18.1	0.8	-15.0	-14.7	1.1	-11.5	11.1	-4.8	9.3	-2.8	2.1	290	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	33.5	20.9	31.7	20.4	30.1	20.0	22.9	29.6	21.9	28.7	21.0	27.5	2.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.9	15.7	25.7	19.9	14.7	24.4	19.0	13.9	23.4	68.1	29.6	64.3	28.7	61.1	27.3	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
8.3	7.1	6.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-19.4	35.9	3.8	2.0	-22.1	37.4	-24.3	38.5	-26.4	39.6	-29.2	41.0		
			-19.6	24.6	3.8	1.3	-22.3	25.5	-24.5	26.3	-26.7	27.0	-29.4	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-2.3	-0.1	5.0	11.5	17.0	20.7	22.6	22.0	16.8	10.8	5.3	-0.5		
	DBStd	9.77	5.24	5.73	4.92	3.91	3.67	3.11	2.96	3.16	3.77	4.29	4.87	5.23		
	HDD10.0	1387	382	283	166	29	2	0	0	0	1	43	154	327		
	HDD18.3	3151	641	516	415	207	68	13	3	6	71	236	390	585		
	CDD10.0	1672	0	0	9	73	219	321	390	373	206	66	14	1		
	CDD18.3	396	0	0	0	2	28	83	135	120	26	2	0	0		
	CDH23.3	3972	0	0	0	36	320	821	1312	1191	270	22	0	0		
	CDH26.7	1483	0	0	0	7	81	280	539	497	76	3	0	0		
(14) Wind	WSAvg	2.5	2.7	2.8	3.1	3.0	2.6	2.4	2.2	2.0	2.2	2.3	2.7	2.6		
(15) Precipitation	PrecAvg	578	31	29	32	48	61	89	75	53	48	35	35	32		
	PrecMax	821	159	66	92	109	206	292	202	162	184	153	112	83		
	PrecMin	371	3	3	2	1	2	12	13	0	0	2	0	0		
	PrecStd	107	24	17	23	27	41	52	46	37	47	30	24	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	15.3	20.9	27.4	31.6	33.2	36.3	36.3	31.5	26.0	20.4	13.2		
		MCWB	6.4	9.6	11.4	16.5	18.8	21.2	20.5	21.3	18.8	17.3	13.8	9.6		
	2%	DB	7.5	11.4	17.8	23.7	28.8	31.4	33.8	33.5	28.9	22.9	16.7	10.0		
		MCWB	5.0	7.3	10.4	14.5	17.9	20.5	20.9	21.0	17.8	15.6	12.5	7.9		
	5%	DB	5.6	8.8	15.0	21.3	26.9	29.8	31.6	31.6	26.5	20.1	14.2	7.7		
		MCWB	3.9	5.9	9.0	13.1	17.3	20.0	20.9	20.4	17.2	14.4	11.1	6.1		
	10%	DB	4.1	6.8	12.6	19.1	24.8	28.1	29.8	29.6	24.1	17.7	12.1	5.7		
		MCWB	2.9	4.6	7.6	12.0	16.5	19.2	20.3	19.9	16.5	13.4	10.1	4.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	10.0	12.5	16.8	20.7	23.6	24.4	24.2	20.6	18.2	14.7	10.2		
		MCDB	9.7	14.1	19.0	25.6	27.6	29.9	30.8	31.4	28.4	24.1	18.9	12.0		
	2%	WB	5.4	7.8	10.8	15.2	19.3	22.2	23.0	22.7	19.2	16.4	13.0	8.1		
		MCDB	7.0	10.4	16.7	22.1	26.3	28.7	29.5	29.8	25.6	20.9	15.9	9.9		
	5%	WB	4.1	6.3	9.5	14.0	18.2	21.1	22.0	21.6	18.1	15.0	11.8	6.1		
		MCDB	5.3	8.4	14.4	19.7	24.8	27.3	28.6	28.6	24.0	19.2	13.9	7.5		
	10%	WB	2.9	4.8	8.1	12.8	17.1	20.1	21.1	20.6	17.2	13.7	10.3	4.4		
		MCDB	4.0	6.6	11.7	17.7	23.1	26.1	27.4	27.2	22.7	17.0	12.0	5.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.7	6.9	9.4	11.1	12.0	11.7	11.9	12.4	11.3	9.7	6.8	5.8		
		MCDBR	7.3	10.3	14.0	15.4	15.1	14.2	15.0	15.6	15.2	14.0	10.3	8.5		
	5% WB	MCWBR	5.4	6.8	7.8	7.4	6.6	5.8	5.3	5.5	6.0	7.4	6.3	6.3		
		MCWBR	6.7	9.6	12.9	13.4	13.5	12.7	12.6	13.6	13.1	12.4	9.1	7.9		
(40) Clear-Sky Solar Irradiance	taub	taub	0.326	0.357	0.392	0.449	0.421	0.418	0.446	0.453	0.410	0.376	0.352	0.323		
		taud	2.265	2.230	2.209	2.083	2.240	2.281	2.209	2.180	2.259	2.333	2.335	2.313		
	Ebn at Noon	Ebn at Noon	736	790	819	803	846	849	817	792	792	762	702	693		
		Edh at Noon	79	102	121	151	134	129	137	135	114	91	75	68		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.20	1.95	3.17	4.44	5.67	5.96	5.96	5.40	3.79	2.42	1.24	0.93		
	RadStd	RadStd	0.17	0.27	0.33	0.39	0.47	0.51	0.45	0.39	0.43	0.29	0.12	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (33 neighbors)	+0.70	N/A	N/A	+0.76	N/A	N/A	-133	-194	+102	+51
		+0.67	N/A	N/A				-144	-200	+101	+37

Nomenclature: See separate page